

CompTIA Server+

Server Hardware Components

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Introduction

Server+

Racking

Hardware Compatibility List (HCL)

Bus Types

Cable Management

Power Connector

Welcome to the **Server Hardware Components** Practice Lab. In this module, you will be provided with information on the hardware components of a Server.

Server Hardware and its form factor are different from desktop computer hardware. Many servers will be using a server rack to store and operate, but there are other types of servers and form factors as well. Cable management will also be another thing to consider. This module will cover some common server components, types of racking, network and power cabling.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Server Components
- Exercise 2 - Racking
- Exercise 3 - Network and Power Cabling
- Exercise 4 - Power Cabling

After completing this module, you should have further knowledge of:

- Hardware Compatibility List (HCL)
- Server Motherboard
- Central Processing Unit (CPU)
- Memory
- Storage
- Bus Types
- Expansion Cards
- Enclosure Sizes and Server Types
- Rack Server Unit Sizes
- Rack Layout
- Redundant Networking
- Twisted Pair
- Fiber Optics Types and Connectors
- Gigabit Ethernet
- 10 Gig Ethernet
- SFP, SFP+ and QSFP
- Cable Management
- Redundant Power Sources
- Power Connector Types

Exam Objectives

The following exam objectives are covered in this module:

1.1 Given a scenario, install physical hardware

- Racking
- Power cabling
- Network cabling
- Server chassis types
- Server components

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

This module contains supporting materials for Server+ (SK0-005).

Click **Next** to proceed to the first exercise.

Exercise 1 - Server Components

Servers have different hardware and software that allow them to function and provide their services. They will have a special operating system that will allow them to choose different services they want to provide and install the necessary software. The hardware will be different because of its purpose. A server is meant to serve many people at one time through simultaneous connections. Also, the hardware for a server is meant to be operational all the time. Stronger, faster processors, bigger motherboards, error-correcting memory, and lots of storage are all common components of servers that differ from desktops.

In this exercise, different server components will be discussed.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Hardware Compatibility List (HCL)
- Server Motherboard
- Central Processing Unit (CPU)
- Memory
- Storage
- Bus Types
- Expansion Cards

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Hardware Compatibility List (HCL)

The hardware compatibility list is a database that tells users what hardware will be compatible with a particular operating system. The list will show common hardware components like video cards, motherboards, audio cards, graphic cards, and other types of hardware and the versions of the operating system it will work with. Referring to these manuals before installing operating systems can validate that the hardware supports it and avoids system issues and failures.

The hardware must be tested with the operating system to build these lists. Companies like Microsoft do thorough testing and documentation to ensure they have a comprehensive list which they then make readily available on their website. Most often, the HCL documents for open-source software will be created and updated by the user community in forums.

Server Motherboard

Servers will have a motherboard that is typically larger than the standard desktop. The motherboard will typically have more than one socket for processors. They can have 12 to 24 RAM slots with an upgraded capacity of 64GB of ECC RAM. High-end server motherboards support memory up to 2TB.

Multiple SATA connections will likely be used for connecting storage drives and larger capacities of drives. Typically, graphics will not be incorporated. Desktop motherboards can house multiple fans to help cool the unit from the extra heat generated by the processors. The motherboard will have a number of expansion slots, network connections, and standard peripheral connections. Server motherboards are usually made on the eATX standard, which is 12"x13" compared to a full desktop at 12"x9.6".

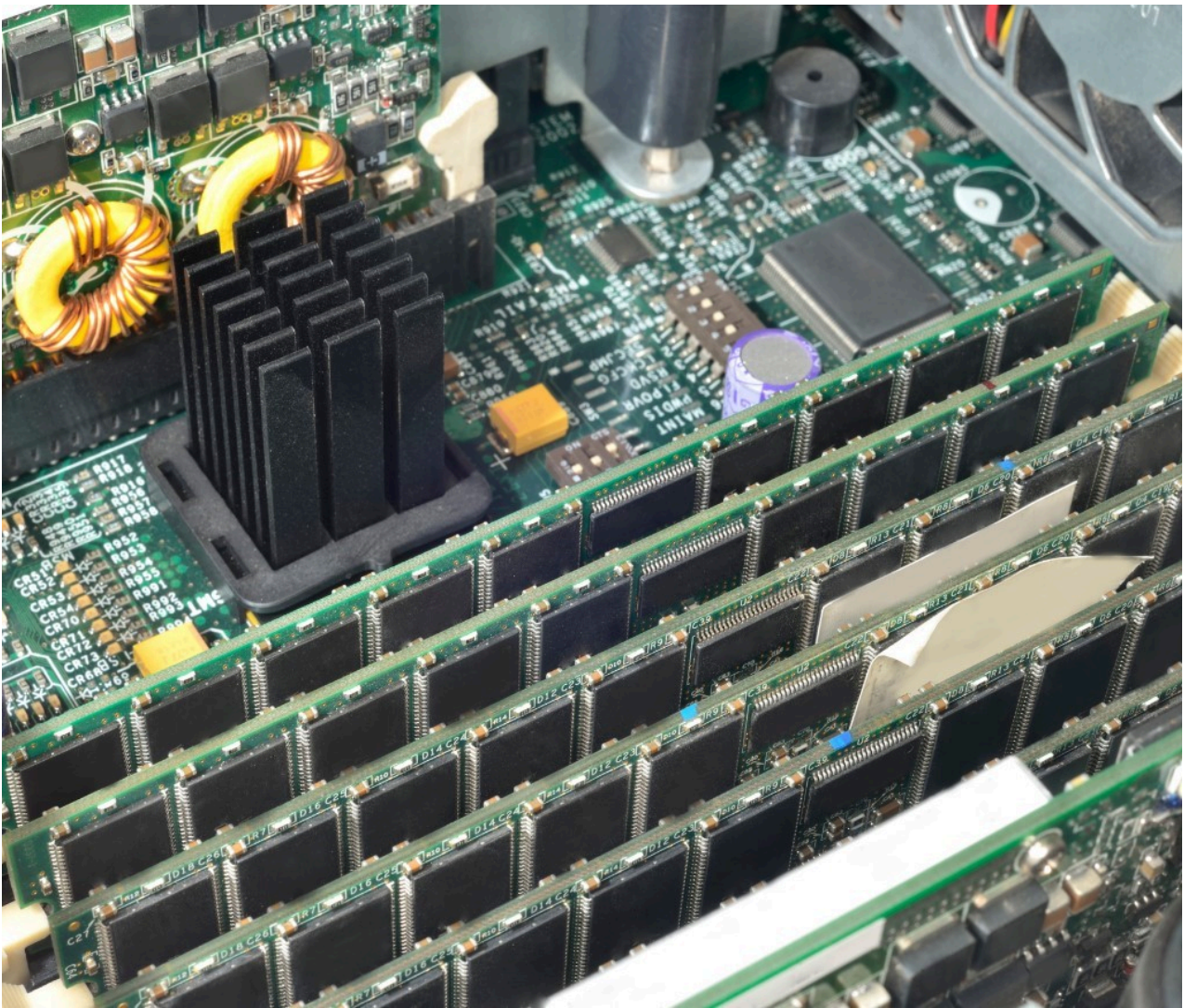


Figure 1.1: Displaying a Server Motherboard.

Central Processing Unit (CPU)

The Central Processing Unit (CPU), also known as the processor, is frequently referred to as the computer's brain, performing the calculations that allow the computers to function. The CPU is inserted into a special socket on the motherboard that will have connections to the different devices and areas on the motherboard. This processor is made up of thousands of transistors that control the flow of electricity to integrated circuits. Along with the transistors and circuits, there is high-performance memory known as cache, which stores data and aids in the loading and unloading of instructions in and out of the processor. All of this must happen in milliseconds, so the processor uses the binary language. A circuit can be turned on, represented by a 1 to the processor or turned off, represented by a 0 to the processor.

The CPU employs binary language to do very quick computations that operate the system, opening and closing programs and performing all the activities that the user is attempting to complete while controlling all the computer's hardware. The image you see on the monitor would not be possible without the processor's instructions. The audio you hear requires a processor. Without CPU input, RAM does not load and release data. The CPU is required for

everything the computer performs, from moving the mouse across the screen to opening a web browser to streaming a video.

Processors have become more powerful now, using less space and resources. CPUs used in desktop and laptop computers differ from those used in servers. The server processor will be quicker and more capable since it will be constructed with more cores and more hyperthreading to handle all the extra processing. Multiple users connect to the servers to access services such as file storage, virtualization, websites, LAN configuration, and others. Depending on the design of the server's motherboard, it may be possible to have more than one CPU. Another distinction is that servers must operate 24 hours a day, seven days a week, which necessitates stronger hardware than workstations, which are supposed to be shut down after a few hours of use each day.

There are two styles of processors common today:

- Pin Grid Array (PGA), produced by AMD, has gold pins on the underside of the processor that plugs into the Zero Insertion Force (ZIF) socket. The processor should fit easily into place, taking care not to bend or break the fragile gold pins. An arm on the side of the socket will be lowered, locking it into place. If there is any resistance, the processor should be taken out and reinserted to ensure it is correct.
- Land Grid Array (LGA), produced by Intel, has flat gold contact points housed on the underside of the processor that rest on the sockets contact points on the motherboard. The processor rests on top of the socket and is secured by a hinged door that covers the edges of the processor and locks it into place. Due to similar performance and less likelihood of the processor being damaged without the pins, AMD has started to employ land grid array for some of its processors.

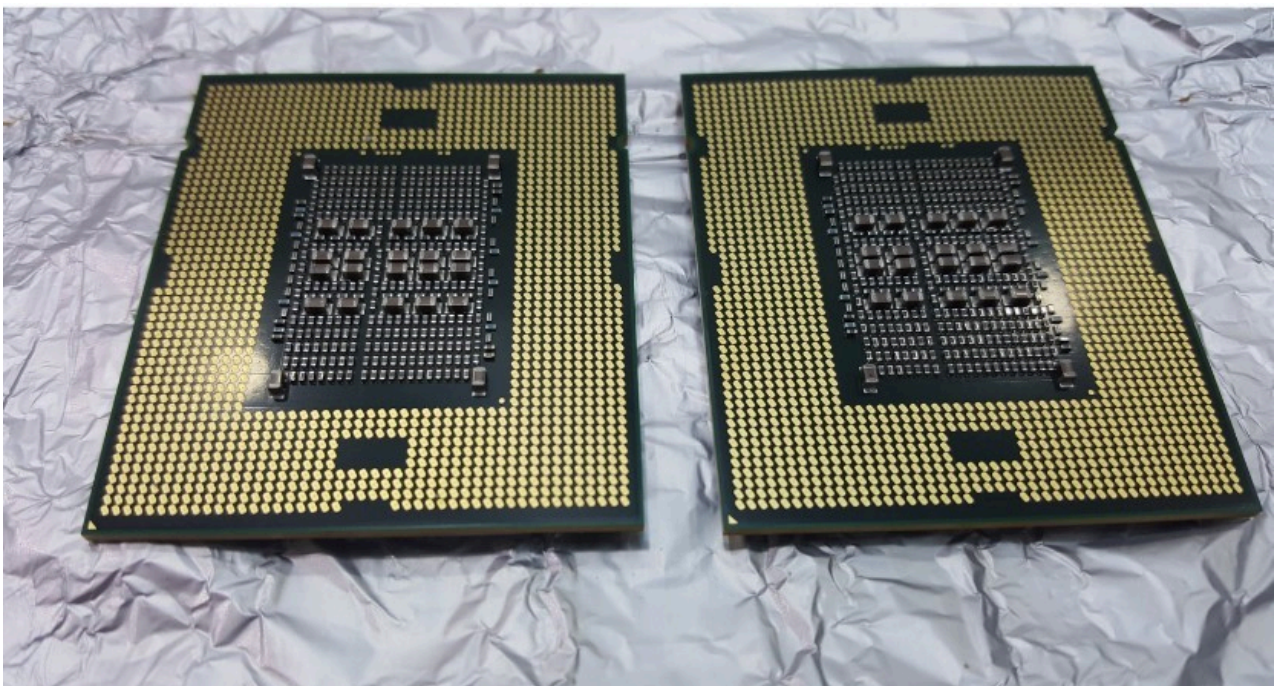


Figure 1.2: Displaying a Land Grid Array (LGA) CPU.

Memory

Servers frequently employ a higher grade of Random Access Memory (RAM) known as Error Correcting Code (ECC). This memory module will have an additional memory chip that validates and corrects any missing information to guarantee that the data is accurately collected and stored. The RAM is regarded as more reliable and stable than standard RAM, but it is significantly more expensive, which is why it is often kept for servers.

ECC is available in both buffered and unbuffered forms. Buffered memory incorporates an additional chip that interfaces directly with the processor, simplifying the operation. Instead of accessing each chip individually to process data which can bog down the processor, as in unbuffered RAM, buffered RAM gives one stream across that extra chip. Because the CPU does not have to remember all the numerous channels and communications that would occur from each chip on the memory stick, it can perform more effectively. Everything is routed through the buffered RAM chip, which controls the RAM module's other RAM chips. Standard servers may not require this level of RAM, so the importance of the server, intended use, desired performance, and cost must all be considered when selecting server RAM.

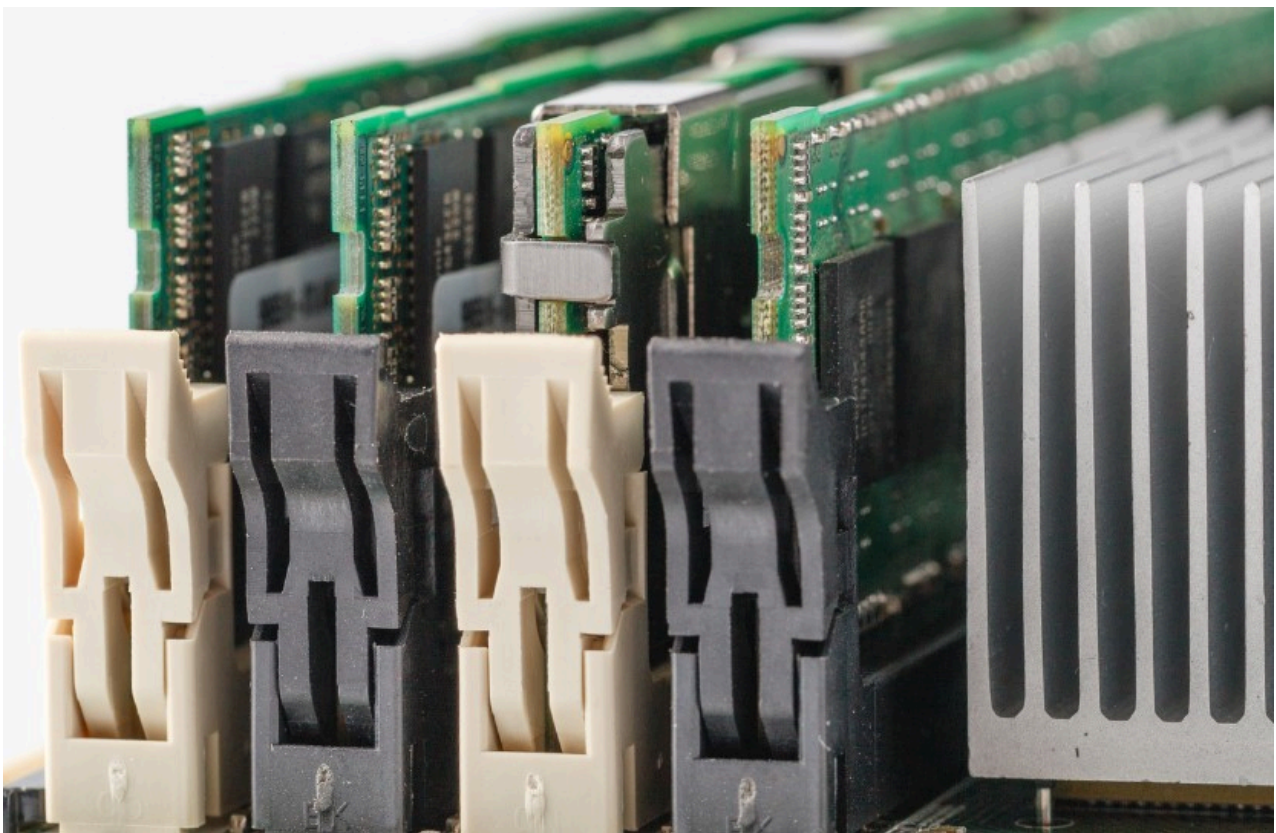


Figure 1.3: Displaying Server Memory modules.

Storage

Servers frequently employ an array of storage drives usually attached through a typical SATA connection. They may be using RAID, or they may manage each disk independently. Larger capacity drives and multiple drives will likely be used to store data for servers.

Depending on the server type, some may require more storage than others. For instance, virtualization servers will host multiple virtual machines through internet connections providing a virtual desktop interface to work with. Each virtual machine will need to store the operating system and the user's data. A DHCP server will be distributing IP addresses. This function requires far fewer resources and storage than a virtualization server.



Figure 1.4: Displaying Server Blades and RAID Storage.



Figure 1.5: Displaying Storage Server in the Cloud.

Bus Types

A bus transports data from one region of the computer to another. This is accomplished by transmitting signals along the motherboard to internal components.

The Northbridge, or frontside bus, directly connects to the CPU and contains the Address Bus, Data Bus, and Control Bus. In a one-way connection, the Address Bus transfers address information of data being stored to the computer's components from the CPU. The Control Bus will transport control signals from the CPU to the components, instructing them on how to operate. This is also a one-way connection, which is done to keep things orderly and to prevent the processor from becoming overburdened while attempting to figure out too much. The Data Bus will be used to transfer data between the components and the processor in both directions. As technology improved, so did the buses, allowing more information to move quickly and efficiently.

The Southbridge, or expansion bus, allows for connecting various peripherals to the computer. The southbridge connects to the northbridge, allowing communication routes on the rest of the board to be added to get the information back to the processor.

Expansion Cards

Peripherals and different interface types are typically attached to the motherboard via expansion cards being plugged into a PCI or PCIe slot connection for services such as audio, video, SCSI storage control, network interface cards, USB connections, RAID Controllers, and a variety of other expansion cards including custom configurations.

Servers will have different external connections that can be made to them. Typically, multiple connections for power supplies, multiple USB slots, serial connections for remote administration, video connections, and the ability to customize and expand with other PCI slots. It depends on the purpose, needs of the server, space, and cost that will determine the build-out and capabilities of the server.

Below are a few common expansion cards.

Graphic Processing Units (GPU)

Graphic Processing Units (GPU) have grown in importance in today's graphic-heavy world. A wide variety of capabilities are available, ranging from built-in graphical processors with minimal capabilities to externally manufactured systems with their own high-powered

specialized processors to deliver high-end graphics. These high-end GPUs feature quicker processors, are more efficient and can compute data other than graphics. GPU Server farms use numerous GPUs to process cryptocurrency, AI learning, Scientific Computing, and Data Analytics. A GPU will not be used with most server configurations, which will most likely be managed by a command line interface like SSH or a KVM to provide graphical input.

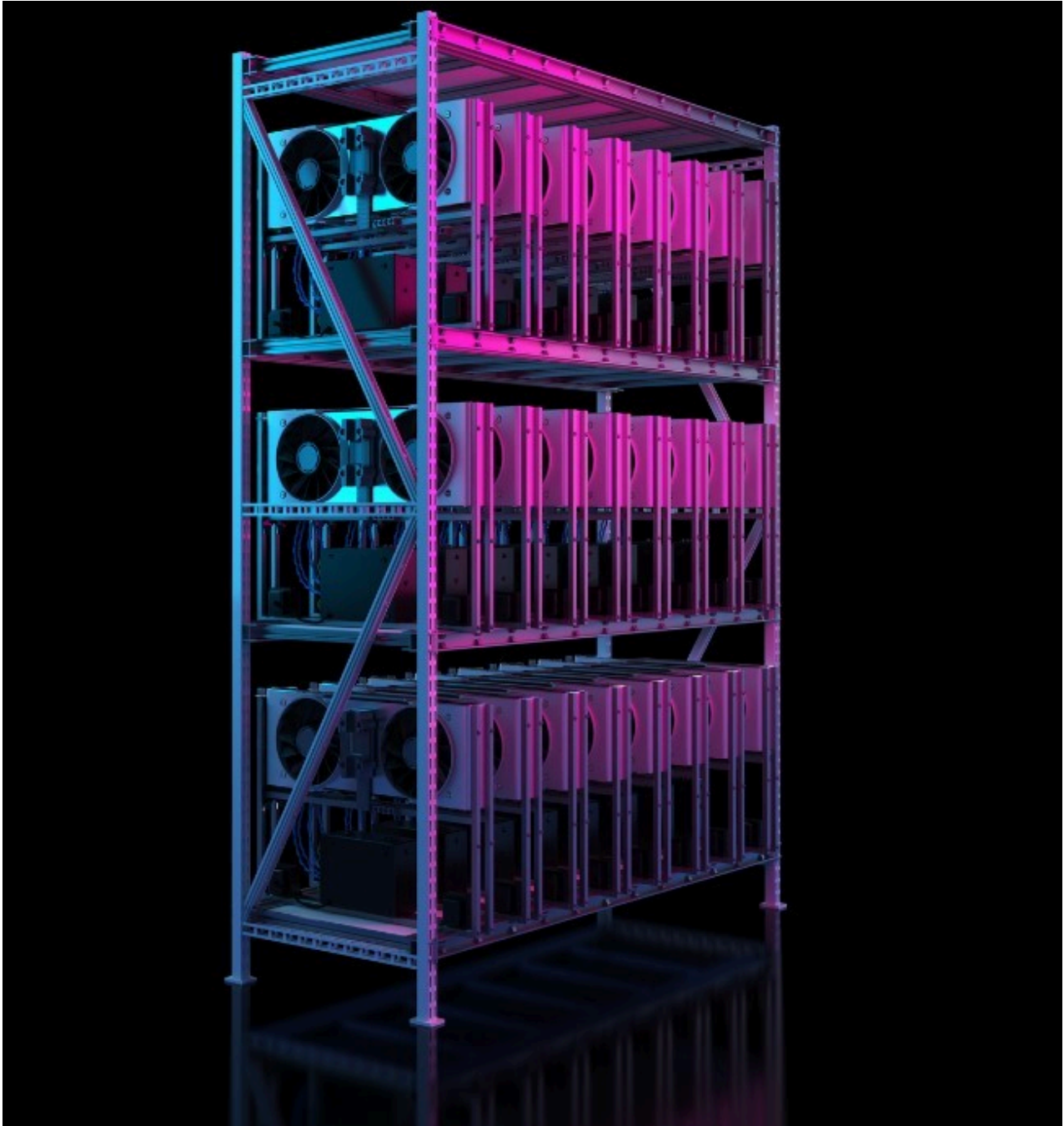


Figure 1.6: Displaying a GPU Rack.

Network Interface Cards (NIC)

Network Interface Cards (NIC) are how the computer and servers connect to the network and send information to other computers, typically through an RJ-45. There are different capabilities, and they have grown through the years of development to provide incredibly stable

and fast connections to wired or wireless connections. These can be ethernet or fiber optics connections. A server will likely have multiple network interface cards to provide redundancy in case of failure.

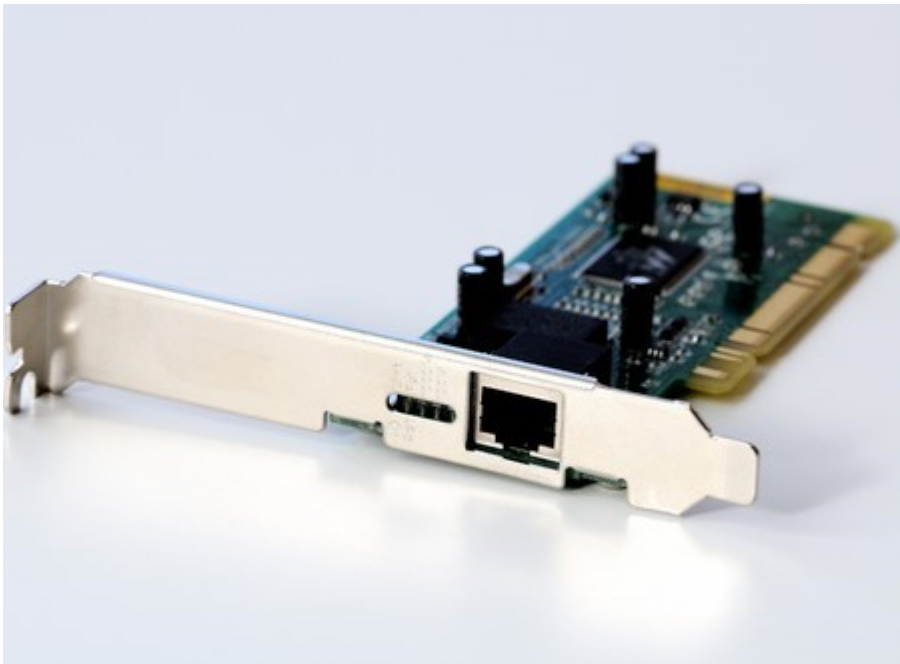


Figure 1.7: Displaying a Network Interface (NIC) Card.

Modem Card

Modem Card will likely be incorporated into the server as an emergency connection to administer the server if the network fails using a command line. This will be done using a dial-up connection from an Internet Service Provider at speeds of 56Kbps.



Figure 1.8: Displaying a Modem Card.

Interface Cards

Interface Cards will add more abilities like additional USB ports, hard drives, specialized ports and other abilities to be added to the computer.

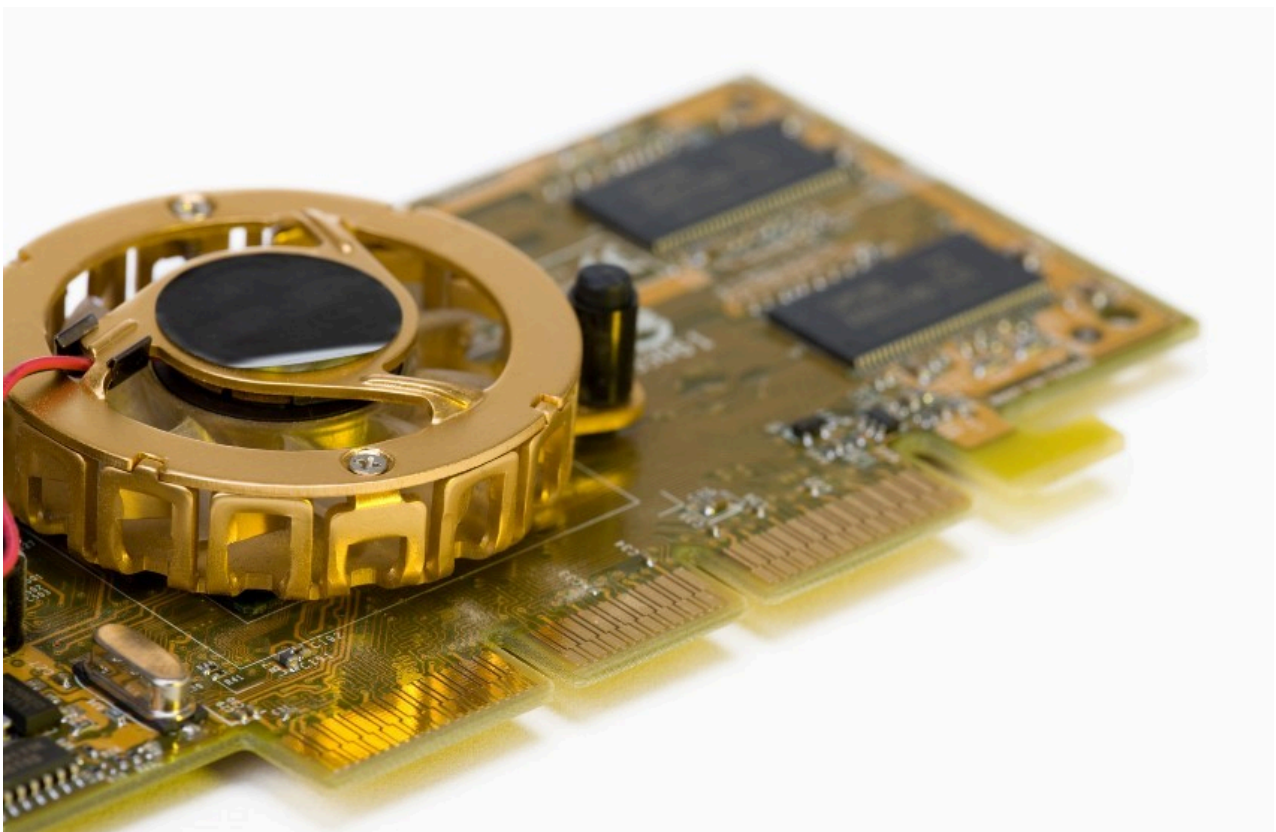


Figure 1.9: Displaying an Interface Card.

Exercise 2 - Racking

Server racking comes in a variety of sizes. When an organization is considering purchasing racking for servers, it should consider the current equipment, available space, and future growth. Data centers will use state-of-the-art racking, whereas a SOHO environment will use smaller racking with fewer features. Each still provides a strong structure to set up the server and has benefits

In this exercise, server enclosure types and general sizes, and rack layouts will be discussed.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Enclosure Sizes and Server Types
- Rack Server Unit Sizes
- Rack Layout

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Enclosure Sizes and Server Types

Tower Server

A Tower Server can be a great addition for small businesses or areas where space is a consideration. The advantage of a tower server over a rack-mounted server is the space-saving **form factor**. A tower server can be as small as a standard full-size desktop or range up to about the size of a filing cabinet. Within the tower will be the CPU, RAM, multiple storage drives, motherboard, cooling and everything else you would expect to see in a standard server.

Tower servers offer a lot of flexibility and allow for easy expansion within their own casing. Once the case is filled, another can be purchased and setup at a relatively low cost and with ease.

The convenience and saving of space do come at some cost. Cabling can be more of a challenge as the form factor increases in size. Putting several tower servers in the same area can be very noisy due to the built-in fans. Typically, they do not have the same computing power as rack-mounted servers but can be configured at a premium cost.



Figure 2.1: Displaying a Tower Server.

Blade Server

A Blade Server is a server that is comprised of “blades” where each blade is a server. A blade is the framework that holds the server’s processor, RAM, storage and input/output ports with an average size of 7” x 19” x 2”. Often the blade will be hot-swappable and slide into the

enclosure and connect to the backplane, which will supply power and connect the blade to the network.

Each blade will be cooled by fans built into the enclosure. Similar to several types of server roles, each blade can be configured to provide a server function. All the blades can work together in cluster mode to provide network resources, share resources for a network, and take part in load balancing to maximize efficiency. These units are small and compact and can fit on top of a desk or be stored within a rack mount. Because each server connects to the backplane and does not need to be wired independently, less wiring is involved. These servers use less power and provide a lot of flexibility.

Some common uses of blade servers include file sharing, web hosting, streaming, load balancing, virtualization, and storage. There are some drawbacks to blade servers, a costly price tag being one of them. This will keep most small businesses from using them and would likely be used in a data center or specialized operations that require computing on a large scale. The backplane that energizes the servers can also become a Single Point of Failure. Although a fan cools each blade, this might not be enough depending on the heat being generated. Additional cooling measures might need to be implemented.



Figure 2.2: Displaying blade servers in a blade enclosure in a rack.

Rack Mounted Server

A Rack Mounted Server is the most common choice for medium to large business use. These enclosures are constructed out of strong metal and are usually tall and slim in stature. They are made to hold rack mounted servers and other networking equipment like routers, switches, patch panels and other networking equipment. This versatility can occur because manufacturers have standardized the sizing of the server. A full rack has a width of 24 inches, a depth of 42 inches, and a height of 73.6 inches tall, and it can carry 42U. Each rack mounted server has a standard height of 1.75,” and this is considered one unit or U. The height of the enclosure will vary depending on how many servers the rack is designed to hold.

These racks make maintenance of the server relatively easy. Servers will be installed on rails that pull out, making it simple to access and work on the server. Using rack mounted servers and enclosures can maximize the space being used because more servers can fit into the space.

A major concern with rack mounted hardware is the dissipation of heat. Although channels for cooling and fans are built into the frame, these are insufficient and additional cooling will likely be necessary. Although there are things in place to make cable management easier, more cabling might be required, which can become overwhelming if not maintained properly. The cost of this enclosure is also significantly higher than other enclosures, but it can often hold more servers.



Figure 2.3: Displaying rack mounted servers.

Rack Server Unit Sizes

The size of servers that go into the rack has been standardized. Servers will have a width of 19 inches and a depth of 29 inches as a standard. This allows for 5" of space between the rack and the server to create airflow and space for wiring. The width of 19" will be the same, but the depth of the server can vary to give additional functions and support for better cable management. The height will go in 1U increments. For example, a 1U server is 1.75 inches. A 2U server will be 3.5 inches. A 4U server will be 7 inches. This means a 4U server will take up 4 server spaces within the rack.

42U and 48U are common sizes for data centers, but the enclosures can get up to 70U in size. When determining what rack is needed, an evaluation of the rack's location, the current

equipment to be used, and foresight for future expansion will need to be considered.

Rack Layout

Cooling Management

Cooling channels are incorporated into the racks themselves. Cold air can be introduced via these channels to reach the machines, while warm air can be extracted from the channels on the opposite side of the rack. Once functional, an infrared camera can be used to spot hot areas of concern so that they can be properly addressed.

Specialized fans the size of a 1U server can be deployed in open spaces to help produce airflow. When numerous racks are deployed, a configuration known as a cold aisle and a hot aisle may be used. The racking will be installed on an elevated platform with air conditioning circulating through vents to the racks. The servers will pull in cooled air through the front and discharge heated air through the back.

Racks of servers will be arranged with their front sides facing one another to form an aisle that will be cooled by air conditioning. The back side of the servers forms the hot aisle, where heat is drawn out of the room. If more than one aisle is created, the back side of the next aisle will face the rear of the first aisle, directing all hot air to the same aisle, while the front side will produce the next cold aisle.

Safety

Rack Balancing and Floor Load Limitations

The layout of the rack is especially important for safety. Larger, heavier servers must be mounted toward the bottom of the rack. Lighter 1U servers and networking equipment should be mounted toward the top of the rack. This will ensure that the rack does not become top-heavy and lessen the likelihood of tipping. Setting the rack up in this fashion will also reduce the risk of injury. Lifting heavy servers and mounting them at higher levels can lead to back injuries and crushing injuries if the server falls on an individual. A 1U server will weigh about 45 pounds. A 42U rack with servers will be roughly 2,000 pounds.

Something that will need to be considered when setting up the racking is the space it will take up and the structure of the flooring. One rack with equipment can reach weights above a ton. The floor must be able to support the weight of the rack. Now imagine multiple racks like there are in a data center. It is crucial to ensure the floor can handle the weight to prevent it from collapsing.

The racks should also be placed in an area of the room away from the air conditioner to prevent any type of water leakage onto the rack and into the server. The servers should also be in an area that is not already hot. Servers produce a lot of heat, so a cooled environment is necessary.

Rail kits will be used to install the server and help with maintenance. You can draw parallels between this and the drawers in a kitchen. The server will be installed on the rails and pulled out like a drawer when the server needs to be accessed. After the maintenance has been finished, the server will be reassembled, and the components will be able to slide back into position. The rails make it easier to access the server and provide a stable platform on which to work. Some larger bottom of the rack servers and equipment, like an uninterruptable power supply, might be fixed at the bottom of the rack with no rails to help prevent uneven weight distribution and tipping.

Proper Lifting Techniques

When installing a server or any other task involving lifting, the technician needs to ensure that they are taking the measures necessary to keep themselves safe. Proper lifting techniques should be followed. The stance should be with feet about shoulder width, establishing a stronger balanced base. The back should be straight throughout the process, and the bend to pick up the item should occur at the knees and hips. With knees bent and a good grip established, slowly start to lift by straightening your legs, ensuring the object stays tight to your body. The further away from the body, the more stress will be placed on the back and neck. Take small steps and use your hips to guide the way. Finally, once reaching the destination, set the object down slowly and controlled.

If an item is too heavy or awkward in size, seek help in lifting the item. Avoid lifting weights above your head. Avoid twisting while supporting the weight. Make sure to utilize any tools available to help, such as a server lift or even back braces to give support while lifting.

Power Distribution Unit

Servers' racks will use something known as a power distribution unit. This hardware solution plugs into the power source and replicates it out of its ports so devices can be plugged in. The setup can be related to a power strip commonly used in homes to give more outlets. Like those power strips, PDUs can help protect against surges to the equipment.

A horizontal PDU will fit inside the rack utilizing 1U or 2U of space. Vertical-mounted PDUs are mounted on the back or side of the rack. Vertical mounted are usually larger PDUs and usually have more outlets than horizontal PDUs. The number of plugs in the PDU will need to

be considered to ensure that the current equipment will all fit and that future growth is being considered.

The electricity and outlets used will be determined based on their geographical location and the electricity standards used there. Some PDUs can have advanced features built into them, like remote management software, environmental monitoring equipment, hot-swap capabilities, reboot devices, auto transfer switching, and individual control of an outlet.

Keyboard, Video, Mouse KVM Placement

The Keyboard Video Mouse, or KVM, is a piece of hardware that utilizes an ethernet or fiber optic connection to link many computers to a single keyboard, display, and mouse. A single button press allows users to switch between any of the linked personal computers. Because they are so simple to operate, they are very useful in data centers and when working with servers. The objective is to minimize the use of space, resources, and financial resources.

By utilizing a single connection to manage a number of computers, businesses can save money by avoiding purchasing separate hardware pieces for each individual computer. The elimination of the requirement to purchase individual monitors for each computer results in a significant reduction in the amount of energy consumed as well as heat dissipated. Additionally, the required quantity of space is decreased. If every server and PC in a data center were equipped with a monitor, keyboard, and mouse, the amount of space needed to house everything would be significantly increased. Several options are available, from pull-out rail kits for the monitors, keyboard and mouse or permanently mounted and used within the rack.



Figure 2.4: Displaying a server with KVM switches.

Exercise 3 - Network and Power Cabling

The types of cables and management of those cables can affect the overall effectiveness of the network. Along with effectiveness, the efficiency and cost of the network can be greatly impacted. A large number of cables can run through a building to supply network access.

In this exercise, you will learn about some common types of cabling and management techniques that will help prevent a tangled mess.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Redundant Networking
- Twisted Pair
- Fiber Optics Types and Connectors
- Gigabit Ethernet
- 10 Gig Ethernet
- SFP, SFP+ and QSFP
- Cable Management

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Redundant Networking

Standard practices dictate that redundancy should be implemented whenever possible to assure compliance with the terms of any Service License Agreements. This encompasses various forms of networking hardware, such as routers, switches, network interface cards, access points, and firewalls, amongst other forms of networking hardware. The goal here is to design your system so that there is not a single point of failure.

The equipment can be in an active/active configuration, which means that each piece is functional and provides a service. Load balancing can be performed where one device is not getting all the traffic, increasing efficiency in the network. It can also be in an active/passive setup where the passive device monitors the active device and takes over if the device fails.

Along with the redundant equipment, there will be redundant wires. All the extra equipment and wires can become very complex to setup and keep running. Different protocols like link aggregation protocol, spanning tree protocol, and other routing protocols may need to be implemented due to the redundant equipment. All this redundancy does cost money. That cost will need to be weighed by the cost of what can happen when the equipment is not functioning and the loss of business.

Twisted Pair

Twisted pair cabling spans all the way back to the invention of the phone. In a twisted pair cable, the signal being sent along the lines will be contained within the wires. These wires are made of copper and have a protective casing around the outside. Pairs of wires are created, and they are twisted around each other. One will be conductive and carry the signal while the other is ground. The twists will be color-coded, and certain colors will be used for certain purposes. Ethernet standard uses Green, Orange, Blue and Brown wires. The twists help to reduce crosstalk, and the more twists, the better the wire will perform.

Unshielded Twisted Pair

Unshielded Twisted Pair will be twisted cabling with the wire and the casing. There is no additional protection to help reduce crosstalk or Electromagnetic interference (EMI). This common wire is used especially when connecting end-user devices to the network. The wires are meant for short distance transmissions and will have attenuation if distances get longer than

100 meters. There is a lower cost to unshielded twisted pairs than shielded twisted pairs and fiber optics.

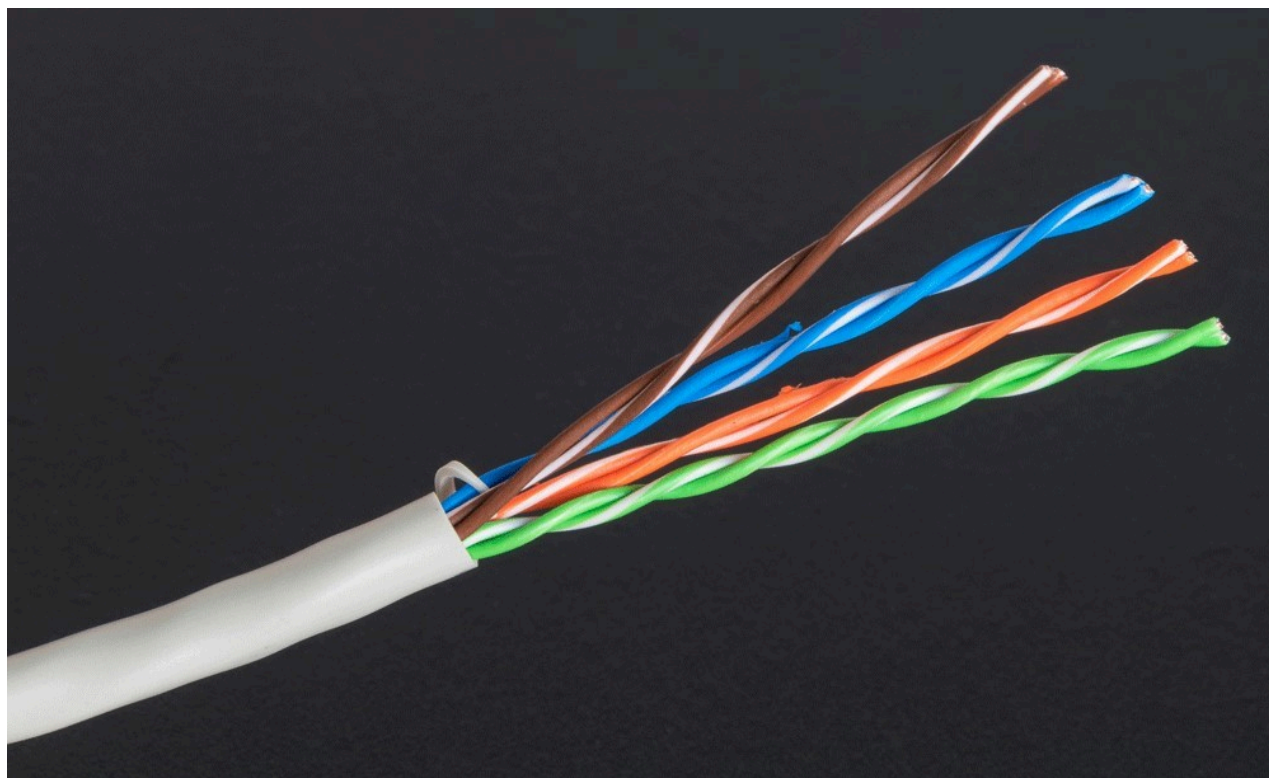


Figure 3.1: Displaying unshielded twisted pair cables.

Shielded Twisted Pair

Shielded Twisted Pair has an additional layer of protection around the wiring. The layer can be a mesh copper casing or a thin foil casing depending on the quality of the wire. Thicker versions of the shielded twisted pair layer add additional structure to the wiring but also makes the wiring more difficult to work with.

Proper grounding of the wires and network is crucial to the network functioning properly. The protective layer helps to protect further from crosstalk and electromagnetic interference. This wire can be used for runs longer in length and perform better than unshielded twisted pairs. The protective layer and additional capabilities make these wires much more expensive than the unshielded twisted pair. There is also a limitation on where they can be used.

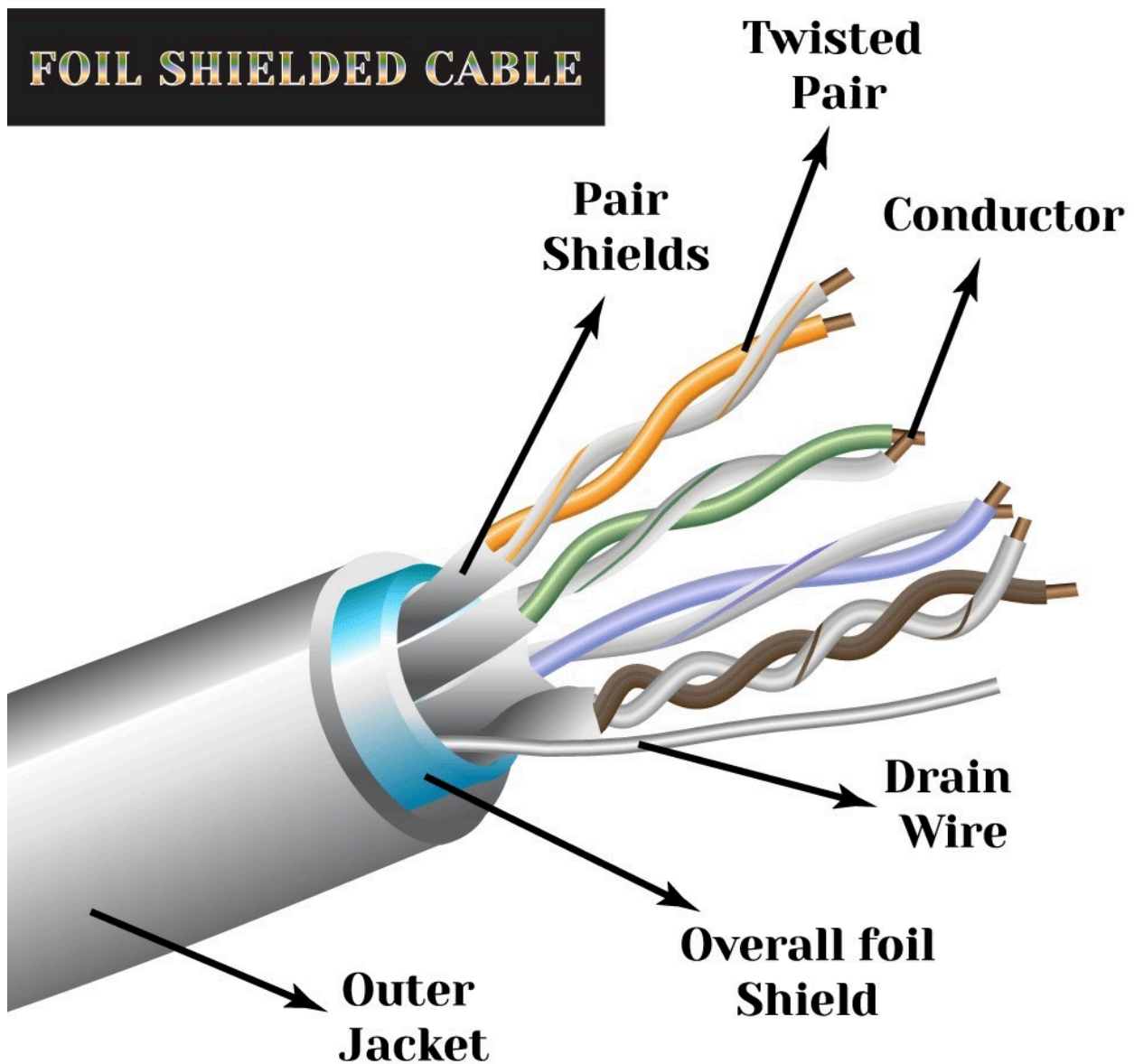


Figure 3.2 Displaying shielded twisted pair cable showing different components of the cable.

Plenum Cables

Plenum Cables are cables with a special coating around them to help if there is a fire in the building. The casing on the wires is non-toxic and will not emit toxic fumes when burned. Also, the coating is self-extinguishing, so it will not continue to feed the fire. These cables will be used between floors and in the plenum space of a drop ceiling. Twisted pair comes in plenum casing as well as other wires.

Fiber Optics Types and Connectors

Fiber optics are used to send and receive data using light as the transmission medium. Unlike twisted pairs, which use metal like copper to send and receive information, fiber optics use a

glass or filament core. A laser or LED light will be used to send the data along the core until it reaches its destination, using light to carry the data.

There are different versions of fiber optic cables. Some are made for longer distances, and some for shorter distances. Fiber optics has increased the speeds and efficiency of networks and has become the backbone of the internet today. In fact, there are undersea cables that connect different continents. Fiber optics can utilize simplex or half duplex design, sending signals along the wire in only one direction. Previously, a duplex could only be achieved by pairing two wires together with a duplex connector. Advancements have been made that now allow for full duplex communication on one fiber optic line using different wavelengths to transmit the data.

Single Mode Fiber Optics

Single Mode Fiber Optics is used when long distances need to be covered, providing the ability to send a signal 6 miles or around 10km before reamplifying the signal. Single mode fiber will have one solid glass core, most likely 9 microns in diameter, running through the center of the cable, protected by layers and cladding to keep the core safe. One thing to pay particular attention to is not bending fiber optics too far to prevent the core from breaking, affecting its ability to transmit light. Once terminated and ready to send data on a line, a laser will be used to transmit the information. Single mode fiber is considered to have limitless bandwidth and is limited to the ability of the transmission and receiving devices. Single mode fiber cables will be labeled with OS.

Multimode Fiber Optics

Multimode Fiber Optics are used for shorter distances and within local area networks because of the attenuation that occurs at longer distances. Gigabit speeds can still be attained at 2km, and 10gb/s can be attained within the 600m mark, so it is a great improvement over today's twisted pair cabling. Multimode fiber has a thicker core than single mode fiber, but the core is split into many different strands, with each transmitting data. The core will likely be made of filament instead of glass and use an LED light to send the signal. Due to the many strands, there is the ability to send different streams of light and data at the same time. Multimode fiber cables will be labeled with OM.

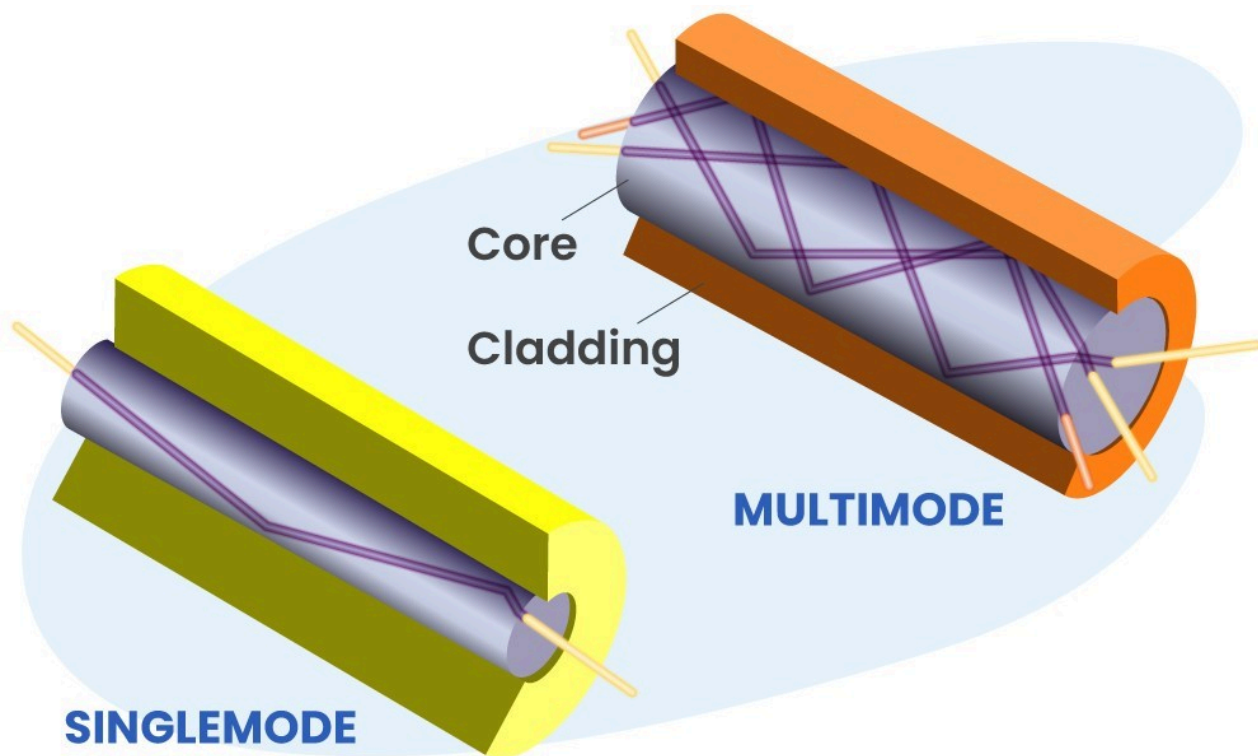


Figure 3.3: Displaying the difference between single mode and multimode fiber cables.

Fiber cables can come with a variety of connectors:

- **LC Connector** - The Lucent Connector is considered the most popular cable type today. This connector is small in stature and is used in many data and telecommunication devices. The connector has a clip and will make a clicking sound when it is connected to the device.

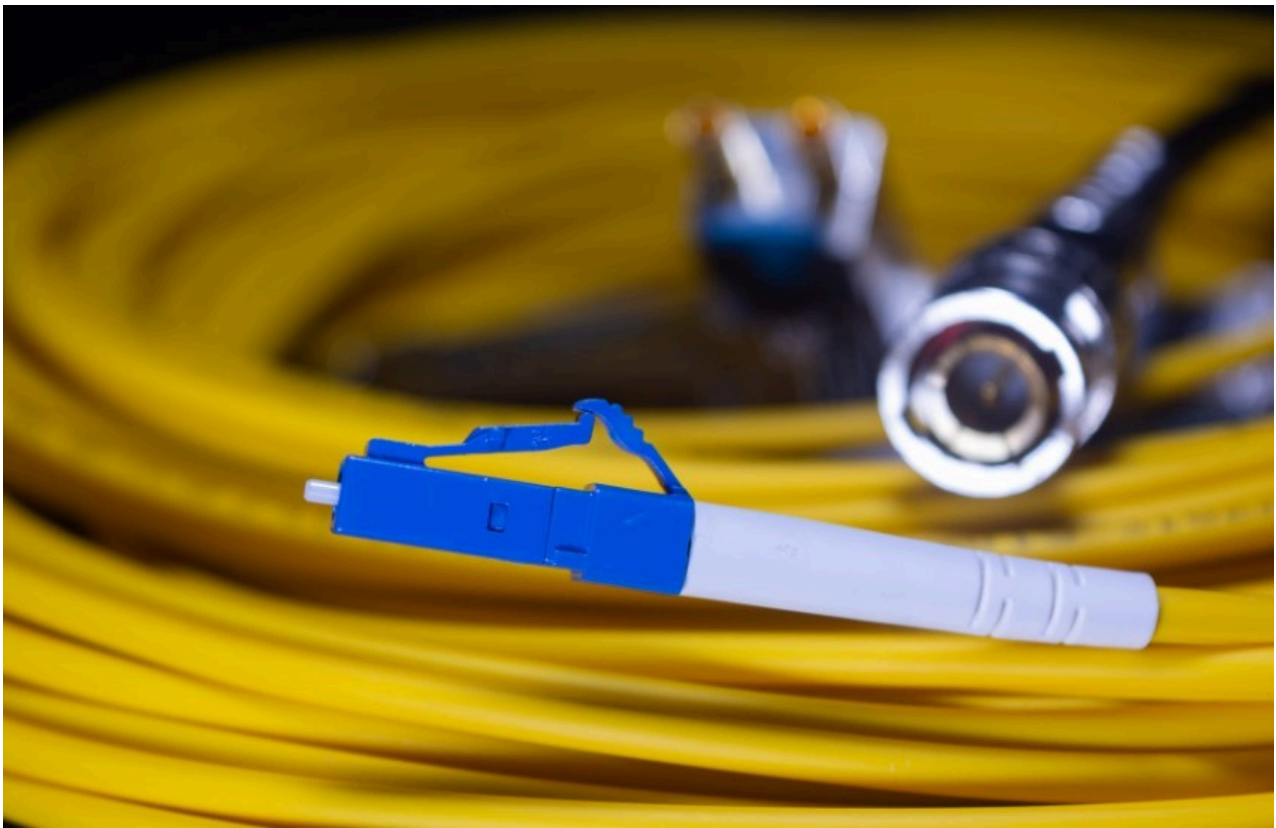


Figure 3.4: Displaying an LC connector.

- **SC Connector** - The Square Connector is considered one of the more popular fiber connectors used today and is frequently used in data and telecommunications. This one does not have any clips or fastening devices and just slides into the housing and snaps into place.

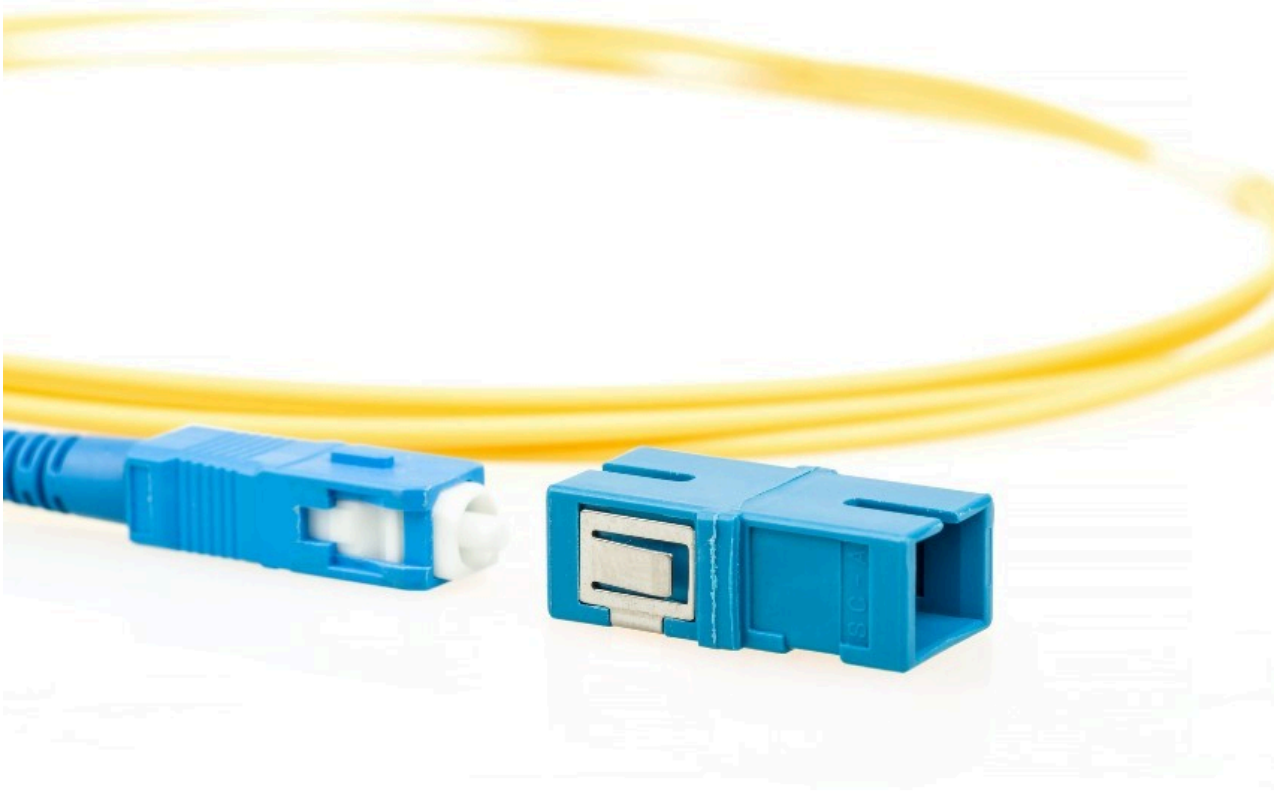


Figure 3.5: Displaying an SC connector.

- **ST Connector** - This is the Straight Tip Connector. It has a collar with a notch cut out that will twist and lock the cable into place. The fiber core is stabilized by a spring-loaded ferrule. This connector is used in both short and long cable runs.

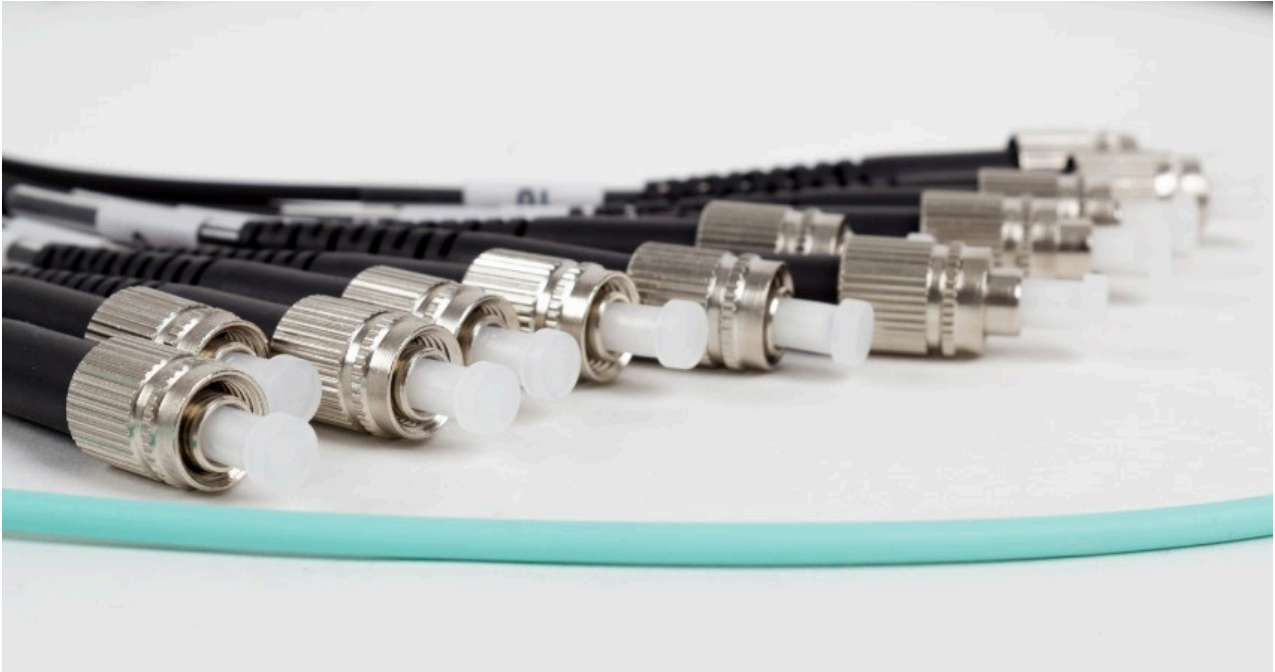


Figure 3.6: Displaying ST connectors.

- **FC Connector** - The Ferrule Connector was one of the original fiber optic connectors. This connector had a screw-type collar paired with a screw-like socket to connect to the device. Proper care has to be taken to ensure the core is aligned correctly.
- **MTP/MPO Connector** - This is a multimode fiber connector combining 12 to 24 fibers into a single connection. The connector will allow for speeds up to 100gb/s. Working with this connector will require some technical expertise.

Gigabit Ethernet

Gigabit Ethernet refers to the cable being able to provide data transfer in excess of 1 GB/s. This type of bandwidth can support many users and user devices connected simultaneously, giving each a high-speed connection to the internet for video streams, music, and business purposes.

The first twisted pair cable that had the capability to do this was Cat5e cabling but could use cable lengths of about 50 meters before attenuation occurs at 100 MHz. Cat6 cables improved upon the technology and can provide speeds of 1 GB/s at the full 100 meters at a 250 MHz frequency. This is all made possible by increasing the number and tightness of the twists within the cable and additional insulation to help prevent crosstalk and EMI.

10 Gig Ethernet

Cat6 also improved on the data transfer speeds reaching 10 GB/s, but like the previous Cat5e cable, that increase could only be sustained for a distance of around 50 meters. Improvements were again made and introduced in Cat6a cabling that will allow the 10 GB/s at the full 100-meter lengths. Thicker cabling, more twists and tighter twists are used to create Cat6a than its predecessor cables. These cables have a much higher cost than Cat5e cabling and will likely be used in businesses and datacenter instead of SOHO environments.

SFP, SFP+ and QSFP

Small Form Factor Pluggable, Small Form Factor Pluggable+ and Quad Small Form Factor Pluggable are hot-swappable transceivers used in ethernet, fiber channel, and other networking communication standards. A hot-swappable transceiver means it can send and receive radio signals and can be plugged and unplugged with continuous power.

The small form factor refers to the networking equipment and connection utilizing the RJ-45 connector. SFP has the capability to reach data transfer rates of 100Mb to 1 GB/s, and SFP+ extends that capability up to 10 GB/s but not less than 1 GB/s. QSFP is another small form factor transceiver with 4 connections, hence the quad in the name. QSFP allows for 4 connections up to 1GB/s. QSFP+ 4x10, QSFP+ 4x14 and QSFP+ 4x28, each number representing the speed capabilities in GB.

Cable Management

Cable management is important to keep things organized. Without cable management, wires can get twisted and crossed and tangled. Now think about what happens when something goes wrong. Finding the wire in the tangle can be like finding a needle in the haystack. Keeping everything organized takes diligence and a few rules that should be followed.

- Label all the wires and cables on both ends of the wire. Label the outlets with the same label so you know where they go. Also, make sure you use the same format to name and label throughout. It may seem like a cumbersome task but once everything is properly labeled, keeping up with new wires is all that is required. When there is an issue, it is much easier to troubleshoot and apply the fix.
- Cabling is not cheap, but less expensive wires will not have the quality of more expensive wires and might break easier or not perform as well. To provide consistent uptime, using cheaper cables might not be the best option. A key to managing costs is making sure you are using the proper length. There should not be too much slack, and it also can't be too tight. Always make sure to measure and measure again to ensure you have it right before cutting. Once the cable is cut, ensure care is taken when attaching it to the connector. Wires need to be stripped to expose the wire. The connector will need to be attached using

a crimper. If attaching an RJ-45 or RJ-11, the crimper will likely have cuts to fit the connector that will clamp down and secure the wires. Some connectors will allow you to enter the wire in without stripping it and will clamp into the wire and cut off any excess wire fed through, making connections easier to terminate.

- When running the wires, watch for pinch points and sharp turns. Bending the wire can influence reliability and performance. This is especially true with fiber optics, where the core will break if bent too far. Cables are usually routed through drop ceilings using hangers. When reaching the servers, wire organization can be incorporated into the rack or may need to be added. With servers being 19 inches wide, the rack leaves some space for cabling and airflow. It is important to distinguish if you are organizing it vertically or horizontally in the rack. Each one will require different measurements and setups.
- Certain cables can experience electromagnetic interference, which can diminish the quality of the network. Separating each wire by types like fiber, ethernet, coaxial, power cables, and other wires can help reduce EMI from happening.
- Do not use zip ties; instead, use velcro straps. To troubleshoot, the zip ties would need to be cut off, so a wire can be cut if not careful. Velcro straps allow you to unfasten and fasten the cable together, making management much better.

Exercise 4 - Power Cabling

Powering the devices is essential and needs to be as redundant as possible. In this exercise, you will learn about common techniques for creating redundancy to ensure constant power to the server.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Redundant Power Sources
- Power Connector Types

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Redundant Power Sources

Servers and other pieces of network equipment can have the capability of having multiple power supplies. In the event that one of the active power supplies fails, the other can step in and continue to supply the required amount of power. When servers are plugged in, they will often be plugged into uninterruptible power supplies (UPSs). After that, they will be connected to an outlet, but it is important to ensure they are not plugged into the same outlet. If one of them stops working, the others should each have their own separate line plugged into the circuit breaker. There is potential for the use of separate circuit breakers and various sources of electrical power coming into the building. After that, to help with uptime, portable generators may be used to take over if the power fails. Standard practices dictate that redundancy should be implemented whenever possible to guarantee the terms of any Service License Agreements.

Power Connector Types

The International Electrotechnical Commission IEC connector is the most common power cable connected to the server in the United States. IEC 13 is used for plugs, and IEC 14 is used for receptacles. National Electrical Manufacturers Association NEMA is the most common cable that plugs into the outlet. Some versions will have a twist lock to ensure the connection is secure, while others will look like standard plugs and come in different amperages.

Review

Well done, you have completed the **Server Hardware Components** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Server Components
- Exercise 2 - Racking
- Exercise 3 - Network and Power Cabling
- Exercise 4 - Power Cabling

You should now have further knowledge of:

- Hardware Compatibility List (HCL)
- Server Motherboard
- Central Processing Unit (CPU)
- Memory

- Storage
- Bus Types
- Expansion Cards
- Enclosure Sizes and Server Types
- Rack Server Unit Sizes
- Rack Layout
- Redundant Networking
- Twisted Pair
- Fiber Optics Types and Connectors
- Gigabit Ethernet
- 10 Gig Ethernet
- SFP, SFP+ and QSFP
- Cable Management
- Redundant Power Sources
- Power Connector Types

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.